

Question 1

Question ID: 1575043

An analyst finds that a particular mutual fund has reported the following annual rates of return:

<u>Year</u>	<u>% Return</u>
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20X1	22
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20X2	-25
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20X3	11
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The fund's geometric mean return over the three-year period is *closest* to:

A) -2.7%.



B) -0.7%.



C) 0.0%.



Explanation

The fund's geometric mean return over the three-year period is approximately -2.74%, calculated as follows:

$$R_G = [(1 + 0.13) \times (1 + 0.18) \times (1 - 0.31)]^{1/3} - 1 = -0.0274 = -2.74\%.$$

(Module 1.1, LOS 1.b)

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Question 2

Question ID: 1575044

Allan Jabber invested \$400 at the beginning of each of the last 12 months in the shares of a mutual fund that paid no dividends. Which method will he correctly choose to calculate his average price per share from the monthly share prices?

A) Arithmetic mean.



B) Harmonic mean.



C) Geometric mean.



Explanation

The harmonic mean of the 12 purchase prices will be his average price paid per share. (Module 1.1, LOS 1.b)

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Question 3

Question ID: 1575045

The EUR/USD spot exchange rate is 0.70145, and one-year interest rates are 3% in EUR and 2% in USD. The forward USD/EUR exchange rate is *closest* to:

A) 1.1426.



B) 1.4118.



C) 1.4396.



Explanation

$$0.70145 \times 1.03 / 1.02 = 0.7083; 1 / 0.7083 = 1.4118.$$

(Module 2.2, LOS 2.c)

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Question 4

Question ID: 1575046

An analyst who wants to display the relationship between two variables graphically is *most likely* to use:

A) a histogram.



B) a scatterplot.



C) a frequency polygon.



Explanation

Scatterplots illustrate the relationship between two variables. Histograms and frequency polygons show the distribution of observations for a single variable. (Module 3.2, LOS 3.d)

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Question 5

Question ID: 1575047

Hamza Abbasi has assembled a portfolio using the following two stocks:

	<u>Stock A</u>	<u>Stock B</u>
Standard deviation	0.12	0.28
Current portfolio weights	0.20	0.80
Expected return	0.08	0.16
Correlation between returns	0.50	

The covariance between the returns on Stock A and Stock B is *closest* to:

A) 0.0168.



B) 0.0657.



C) 0.2143.



Explanation

The correlation between two stock returns can be expressed as follows:

$$\rho(R_i, R_j) = \text{Cov}(R_i, R_j) / [\sigma(R_i) \times \sigma(R_j)]$$

Rearranging, the covariance can be calculated as:

$$\text{Cov}(R_i, R_j) = \rho(R_i, R_j) \times \sigma(R_i) \times \sigma(R_j)$$

For Stocks A and B, the covariance is:

$$\text{Cov}(R_A, R_B) = \rho(R_A, R_B) \times \sigma(R_A) \times \sigma(R_B) = 0.50 \times 0.12 \times 0.28 = 0.0168.$$

(Module 5.1, LOS 5.a)

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Question 6

Question ID: 1575048

All else being equal, as the correlation between two assets approaches -1.0 , the diversification benefits:

A) increase.



B) stay the same.



C) decrease.



Explanation

Diversification benefits increase as two assets approach perfect negative correlation. In other words, the less positively correlated assets are, the greater the advantage is from diversification.

(Module 5.1, LOS 5.a)

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Question 7

Question ID: 1575049

If the continuously compounded daily return for Fittizio S.p.A. shares follow a normal distribution, the probability distribution for Fittizio share prices is *most likely* to follow a:

A) normal distribution.



B) Bernoulli distribution.



C) lognormal distribution.



Explanation

Fittizio share prices should follow a lognormal distribution if the continuously compounded daily returns on Fittizio shares follow a normal distribution.

(Module 6.1, LOS 6.a)

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Question 8

Question ID: 1575050

The percentage changes in annual earnings for a company are approximately normally distributed with a mean of 5% and a standard deviation of 12%. The probability that the average change in earnings over the next five years will be greater than 15.5% is *closest* to:

A) 2.5%.



B) 5.0%.



C) 10.0%.



Explanation

The standard error of a 5-year average of earnings changes is $\frac{12\%}{\sqrt{5}} = 5.366\%$. 15.5% is $\frac{15.5-5}{5.366} = 1.96$ standard errors above the mean, and the probability of a 5-year average more than 1.96 standard errors above the mean is 2.5% for a normal distribution. (Module 7.1, LOS 7.b)

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Question 9

Question ID: 1575051

An analyst wants to construct a hypothesis test to determine whether the mean weekly return on a stock is positive. The null hypothesis for this test should be that the mean return is:

A) greater than zero.



B) less than or equal to zero.



C) greater than or equal to zero.



Explanation

The null hypothesis should state the condition which, if rejected, would lend statistical support to the alternative hypothesis. Here, the null hypothesis should be that the mean return is less than or equal to zero, and the alternative hypothesis should be that the mean return is greater than zero. (Module 8.1, LOS 8.a)

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Question 10

Question ID: 1575052

An analyst should use a t -test with $n - 1$ degrees of freedom for a test of:

A) mean differences.



B) equal variances.



C) correlation.



Explanation

Mean-differences hypothesis tests are t -tests with $n - 1$ degrees of freedom. Tests of correlation are t -tests with $n - 2$ degrees of freedom. Tests for equality of variances are F -tests. (Module 8.1, LOS 8.a)

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Question 11

Question ID: 1575053

A researcher estimates a simple linear regression for two variables using 47 observations. The regression sum of squares is 0.0182 and the sum of squared errors is 0.0375. The coefficient of determination for the regression is *closest* to:

A) 32.7%.



B) 48.5%.



C) 67.5%.



Explanation

The coefficient of determination = $0.0182 / (0.0182 + 0.0375) = 32.7\%$. (Module 10.2, LOS 10.c)

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Question 12

Question ID: 1575054

Neural networks are an example of:

A) machine learning.



B) artificial intelligence.



C) algorithmic trading applications.



Explanation

Artificial intelligence refers to systems that can be programmed to simulate human cognition. Neural networks are one example of this type of system. (Module 11.1, LOS 11.b)

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